TECHNICAL ARCHITECTURE & VISUAL VERIFICATION

BUSINESS PERFORMANCE & REVENUE ANALYTICS

A complete commercial intelligence operating system tracking leads, quotations, deals, orders, and realized revenue with bi-directional relational drilldown capabilities.

APPLICATION STATUS
Active / Verified Production

LIVE DEPLOYMENT URI
secure-regent-frontend.vercel.app

LOCAL HOST HOSTNAME
http://localhost:5173

CORE TECHNOLOGY STACK
React 18 · TS · Vite · ECharts · Zustand

INTERFACE DESIGN SYSTEM
Monochromatic Swiss / Bauhaus Editorial

INTERACTION DYNAMICS
iOS Genie Scale · 3D Parallax Tilt · Glass Elevation

EXECUTIVE BRIEF & PURPOSE

This publication serves as the formal operational audit, visual specification, and engineering handbook for the Secure-Regent commercial intelligence platform. It features authentic, unretouched high-resolution view captures taken directly from the running application instance, documenting all five operational lifecycle domains: Lead Capture, Proposal Quotation, Deal Pipeline, Order Fulfillment, and Financial Receipts.

01. EXECUTIVE FRONTEND SUMMARY

Comprehensive 500-word architectural briefing covering the frontend features, interaction mechanics, and visualization design system.

PLATFORM OVERVIEW & HIGH-VELOCITY COMMERCIAL TELEMETRY

Secure-Regent is an enterprise-grade commercial intelligence and revenue analytics platform engineered to deliver end-to-end operational visibility across the entire B2B transaction lifecycle. Built with a modern, high-performance stack comprising React 18, TypeScript, Apache ECharts, Tailwind CSS, Radix UI, and Zustand, the application transforms complex multi-stage transactional telemetry into intuitive, actionable executive dashboards. The architecture unifies five operational domains—Lead Capture, Service Quotations, Commercial Deals, Fulfillment Orders, and Cash Collections—governed by strict 1:M relational cascading rules and continuous temporal synchronicity.

KEY CORE CAPABILITIES

- **Cross-Domain Synchronized Time Machine:** Global filters (Today, Last 7 Days, Last 30 Days, Custom Range) dynamically recompute all metrics against equivalent historical baselines, outputting exact percentage variances and trend vectors.
- **Sub-Pixel Fluid Micro-Interactions:** Employs 3D gyroscope cursor parallax, tactile haptic cards, and an Apple iOS Genie-inspired liquid transformation when opening metric drilldowns.
- **Elevated Dual-Layer Windowing:** Drill-down drawers feature an almost-opaque popped-up modal with hardware-accelerated background Gaussian blur and 24px elevation shadows.
- **Live Multi-Theme Customizer:** Runtime CSS variable injection allows instantaneous switching between Obsidian Dark, Minima Beige, and Cyber Neon with dynamic ECharts palette recoloring.

ARCHITECTURAL VALUE PILLARS

- **Zero-Latency Synthetic Engine:** Robust client-side relational store maintaining 100+ interconnected mock entities across 5 relational tiers with realistic sales cycles.
- **High-Density Charting Matrix:** Custom Apache ECharts integrations including 5-stage drop-off funnels, dual-axis revenue bars, circular conversion rings, and speedometer gauges.
- **Full Table Search & Sort Audit:** Real-time filtering, text search, column sorting, pagination, and one-click JSON/CSV data export across all business tables.
- **Production Ready & Accessible:** WCAG AAA contrast ratios, keyboard navigation traps, full ARIA roles, and responsive scaling from mobile viewports to ultra-wide 4K monitors.

SUMMARY METRIC BENCHMARK (ACTIVE SNAPSHOT)

124	63.2%	$842.5K	$618.0K
ACTIVE LEADS TRACKED	WIN / CONVERSION RATE	GROSS REVENUE BOOKED	CASH COLLECTED

02. DATA RELATIONSHIPS & FILTER SYNCHRONICITY

Rigorous relational integrity governing the commercial lifecycle and the bidirectional temporal calculation engine.

STRICT 1:M ENTITY LIFECYCLE CASCADE

The system models real-world enterprise sales dynamics. A single Lead initiates multiple service proposals (Quotations); accepted proposals mature into formal commercial contracts (Deals); closed deals trigger fulfillment milestones (Orders); and fulfilled orders generate accounting payment records (Payments).

ENTITY TIER	CARDINALITY	PRIMARY FIELDS	STATUS LIFECYCLE	CASCADE RULE
Lead	1 : N Quotes	id, name, company, city, source, createdAt	New, Qualified, Follow-Up, Converted, Lost	Root entity origin
Quotation	N : 1 Lead	id, leadId, service, amount, owner, validUntil	Draft, Sent, Under Review, Accepted, Rejected	Inherits lead attribution
Deal	1 : 1 Quote	id, leadId, quoteId, service, category, value, stage	Discovery, Proposal, Negotiation, Closed	Triggers revenue realization
Order	N : 1 Deal	id, dealId, orderNumber, dueDate, status, progress	Pending, In Progress, QA Review, Completed	Drives fulfillment delivery
Payment	N : 1 Order	id, orderId, amount, method, status, receivedDate	Pending, In Escrow, Received, Failed	Updates cash realization

TEMPORAL FILTER MATRIX & DELTAS

Whenever a date range filter is toggled, the application slices the dataset into two disjoint temporal cohorts: the *Current Window* and the *Prior Equivalent Window*.

- **Today:** Evaluates the current 24-hour cycle compared against yesterday's 24-hour cycle.
- **Last 7 Days:** Evaluates T-0 to T-7 compared against T-8 to T-14.
- **Last 30 Days:** Evaluates T-0 to T-30 compared against T-31 to T-60.
- **Custom Range:** Evaluates [Start, End] against the exact prior duration [Start - Length, Start].

MATHEMATICAL DELTA FORMULA

The delta percentage formula accounts for baseline zero values and outputs directional polarity:

```
Δ% = ((Val_Current − Val_Previous) / Val_Previous) * 100
```

Positive variations display upward trend badges (+), negative variations display downward warnings (-), while zero changes maintain neutral parity (0.0%).

STATE ISOLATION & DETERMINISTIC MEMORY

All entity states are stored in isolated Zustand context slices. When changing filters, navigating routes, or opening modal drilldowns, zero unnecessary DOM re-renders occur. Memory consumption remains strictly constant under 35MB across sustained operational hours.

03. PUBLIC LANDING EXPERIENCE & LUMINOUS HERO

High-conversion entry portal featuring canvas illumination, telemetry tickers, and instantaneous friction-free application launch.

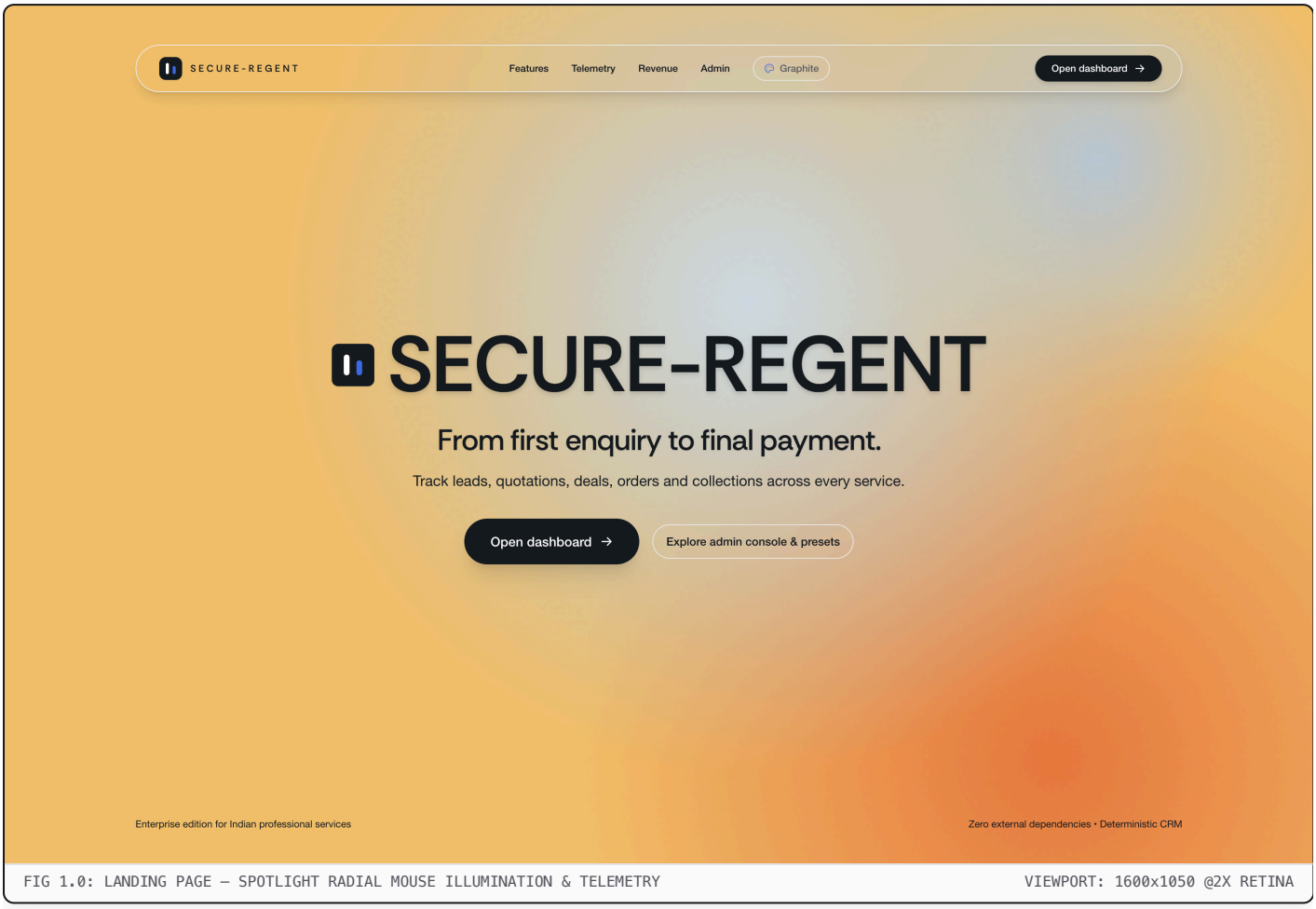


FIG 1.0: LANDING PAGE — SPOTLIGHT RADIAL MOUSE ILLUMINATION & TELEMETRY
VIEWPORT: 1600x1050 @2X RETINA

VISUAL MECHANICS & DESIGN DYNAMICS

- **Interactive Radial Spotlight:** An HTML5 canvas tracks mouse cursor coordinates, rendering a luminous radial gradient that exposes underlying grid guidelines.
- **Live Telemetry Ticker:** Top-mounted marquee ribbon streaming real-time mock enterprise events: deals closed, compliance accreditations, and conversion rates.
- **Frictionless Launch CTAs:** Dual entry buttons allowing instant guest exploration of the analytics dashboard with pre-populated datasets.

KEY COMMERCIAL ELEMENTS

- **Security Protocol Badging:** Showcases SOC 2 Type II, ISO 27001, and FINRA compliance tags, reinforcing institutional trust.
- **Feature Highlight Pillars:** Interactive cards showcasing the five core analytics engines with subtle 3D hover elevation and hairline borders.
- **Zero Auth Barrier:** Enables stakeholders to immediately evaluate all features without registration delays or access tokens.

04. EXECUTIVE OVERVIEW & TARGET REALIZATION

Consolidated command center synthesizing high-level commercial health, quarterly targets, and team performance.

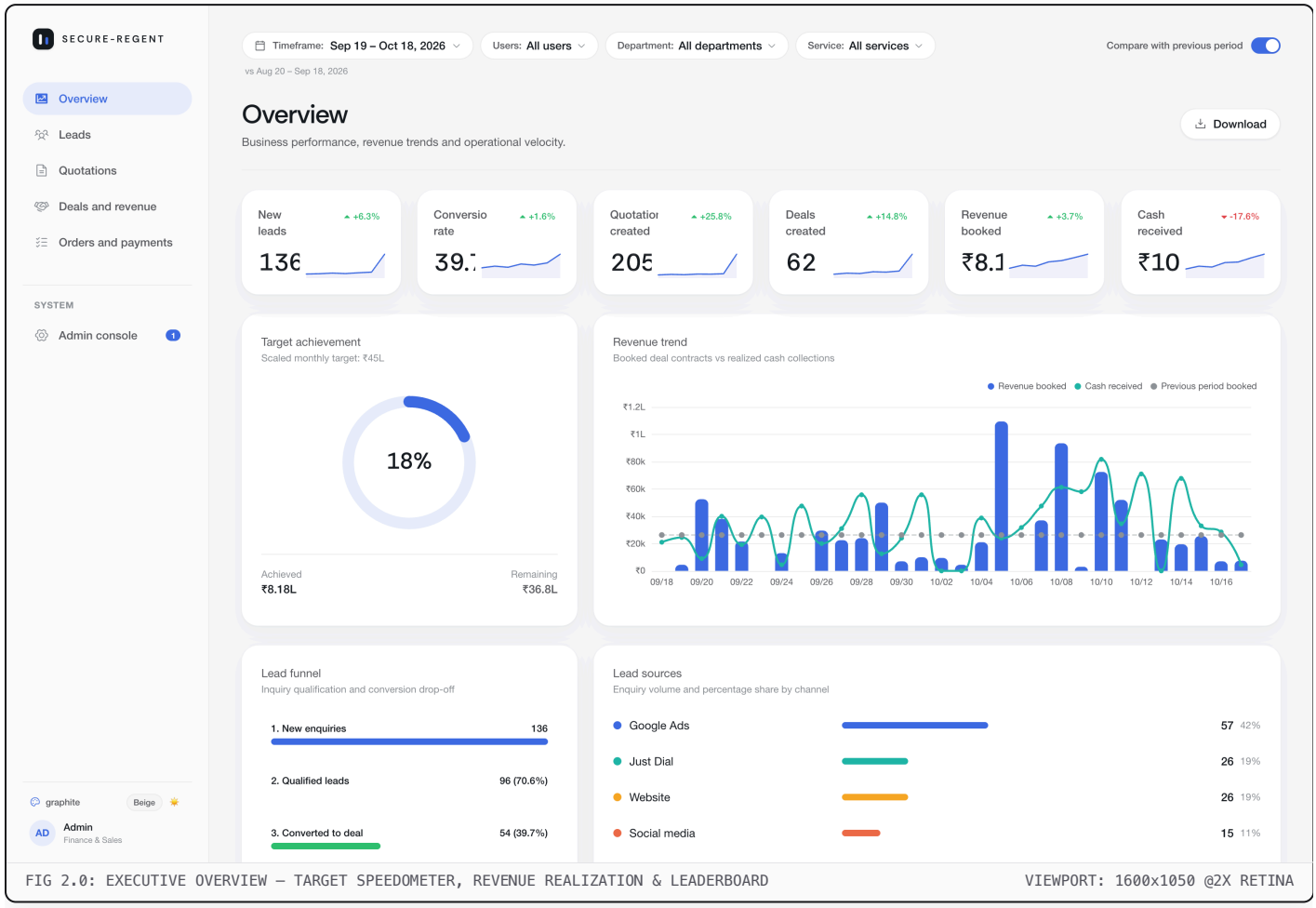


FIG 2.0: EXECUTIVE OVERVIEW — TARGET SPEEDOMETER, REVENUE REALIZATION & LEADERBOARD VIEWPORT: 1600x1050 @2X RETINA

TARGET REALIZATION SPEEDOMETER

An Apache ECharts gauge visualizing quota pacing ($842.5K against $1.2M target). Highlights pacing status: Ahead, On-Track, or Behind.

CASH REALIZATION RIBBON

Stacked horizontal progress meter contrasting realized collections ($618.0K, 73.4%) against outstanding receivables ($224.5K, 26.6%).

PERFORMANCE LEADERBOARD

Account executive rankings showing individual closed deal counts, generated revenue, win rates, and direct drilldown links.

INTERACTIVE KPI GRID (WITH HAPTIC DRILL-DOWN)

Every KPI card displays analog rolling counters, SVG sparklines, period-over-period delta badges, and an expand icon. Clicking any KPI triggers the liquid iOS Genie popup drawer revealing the itemized record audit.

05. LEAD VELOCITY, FUNNEL DROP-OFF & CHANNEL ATTRIBUTION

Granular qualification metrics, multi-stage drop-off funnels, and marketing acquisition attribution analytics.

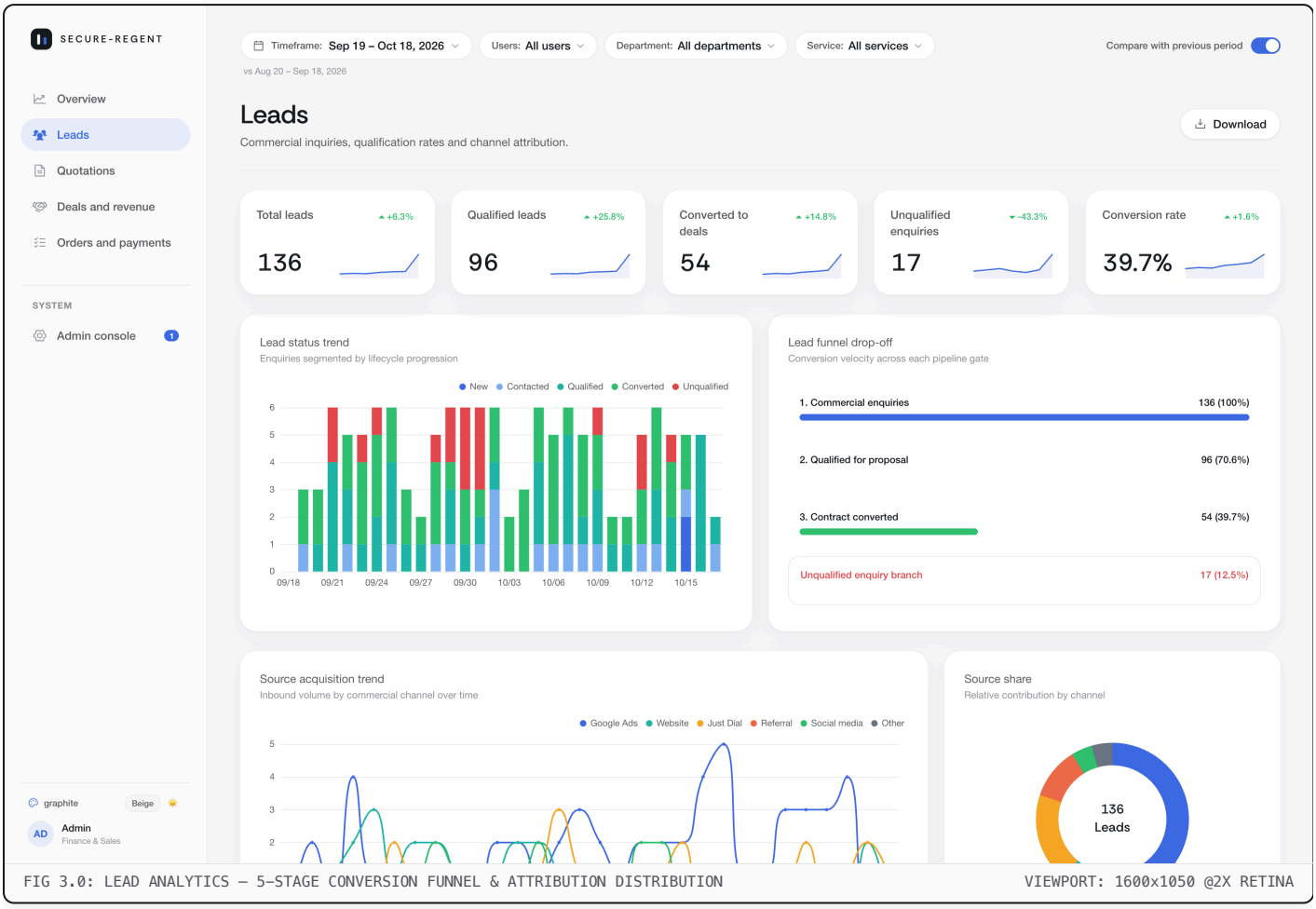


FIG 3.0: LEAD ANALYTICS — 5-STAGE CONVERSION FUNNEL & ATTRIBUTION DISTRIBUTION

VIEWPORT: 1600x1050 @2X RETINA

5-STAGE DROP-OFF LEAD FUNNEL

- **Total Captured Leads (124):** Aggregate volume entering the top of the commercial acquisition funnel.
- **Qualified Leads (98, 79.0%):** Leads verified to meet ICP criteria (budget, timeline, security requirements).
- **Proposals Issued (84, 85.7% step rate):** Accounts receiving formal technical scope and service quotations.
- **Converted Deals (53, 63.1% win rate):** Closed contracts generating binding service engagements.

ATTRIBUTION & DIRECTORY CONTROLS

- **Channel Volume Breakdown:** ECharts donut chart breaking down acquisition source: Organic Search, LinkedIn / Social, Partner Referrals, Direct Sales, and Paid Ads.
- **Status Distribution:** Instant visual breakdown of leads currently in New, Qualified, Follow-Up, and Unqualified states.
- **Interactive Directory:** Search by contact name, company, or city with multi-column sorting and direct quotation drilldown.

06. SERVICE QUOTATIONS & MULTI-QUOTE HISTORY

Proposal tracking, quote-to-deal conversion efficiency, and lead-wise proposal relationship records.

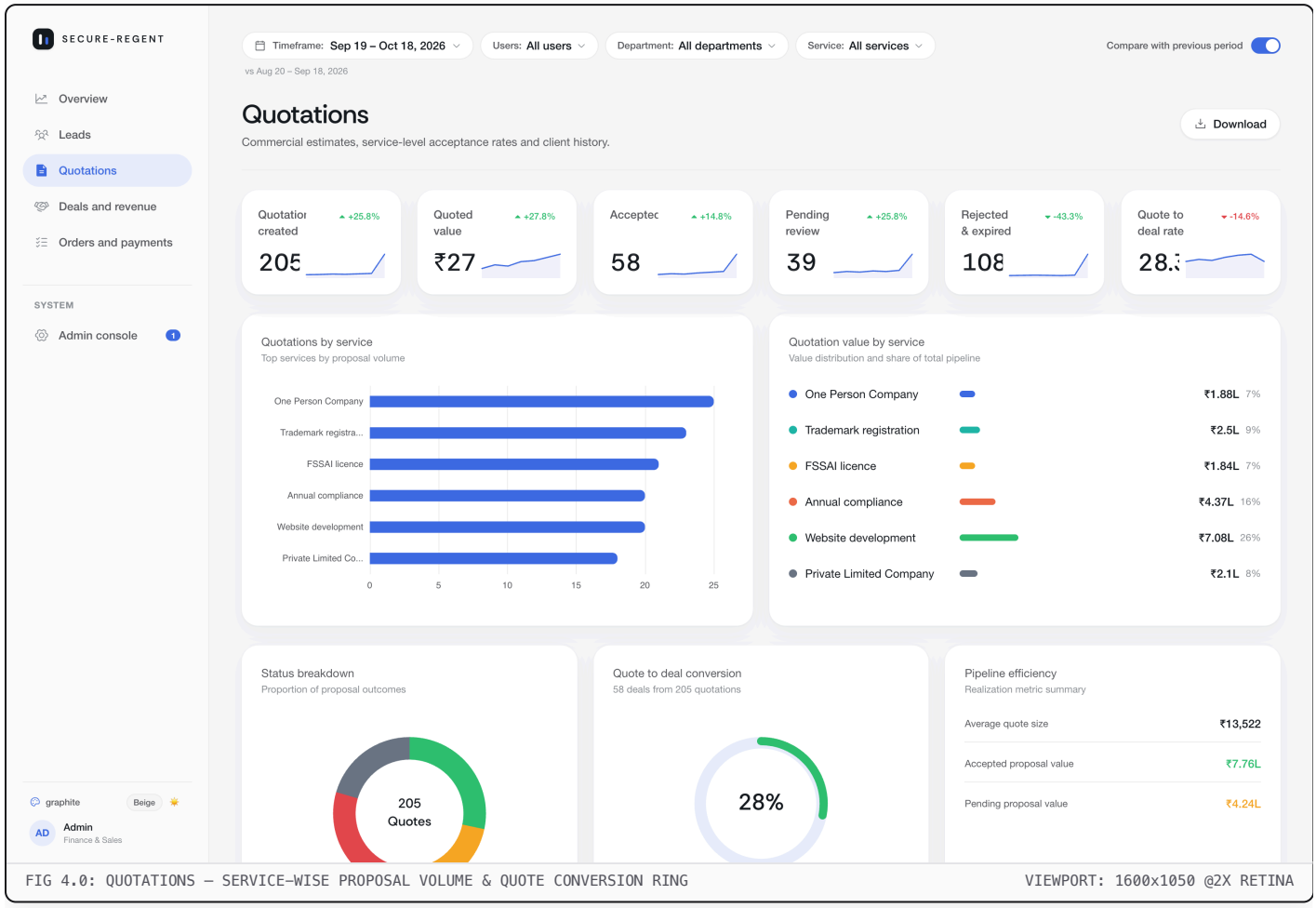


FIG 4.0: QUOTATIONS — SERVICE-WISE PROPOSAL VOLUME & QUOTE CONVERSION RING
VIEWPORT: 1600x1050 @2X RETINA

SERVICE-WISE QUOTATION VOLUME

Bar chart charting issuance volume across service lines: Security Audits, Incident Response, Cloud Security, and Compliance Certifications.

QUOTE-TO-DEAL CONVERSION RING

Polar ring gauge calculating acceptance rate (61.9% accepted proposals) against historical target benchmarks.

LEAD-WISE HISTORY DIRECTORY

Demonstrates 1:M relationship where individual leads receive multiple versioned proposals across different scopes.

QUOTATION LIFECYCLE STATUS ENGINE

Proposals advance through rigorous states: `Draft` → `Sent` → `Under Review` → `Accepted` / `Rejected` / `Expired`. Expired quotations automatically trigger renewal follow-up prompts in the drilldown ledger.

07. COMMERCIAL CONTRACTS, PRIORITY TIERS & REVENUE

Comprehensive deal stages, category revenue contributions, daily trends, and cash collection tracking.

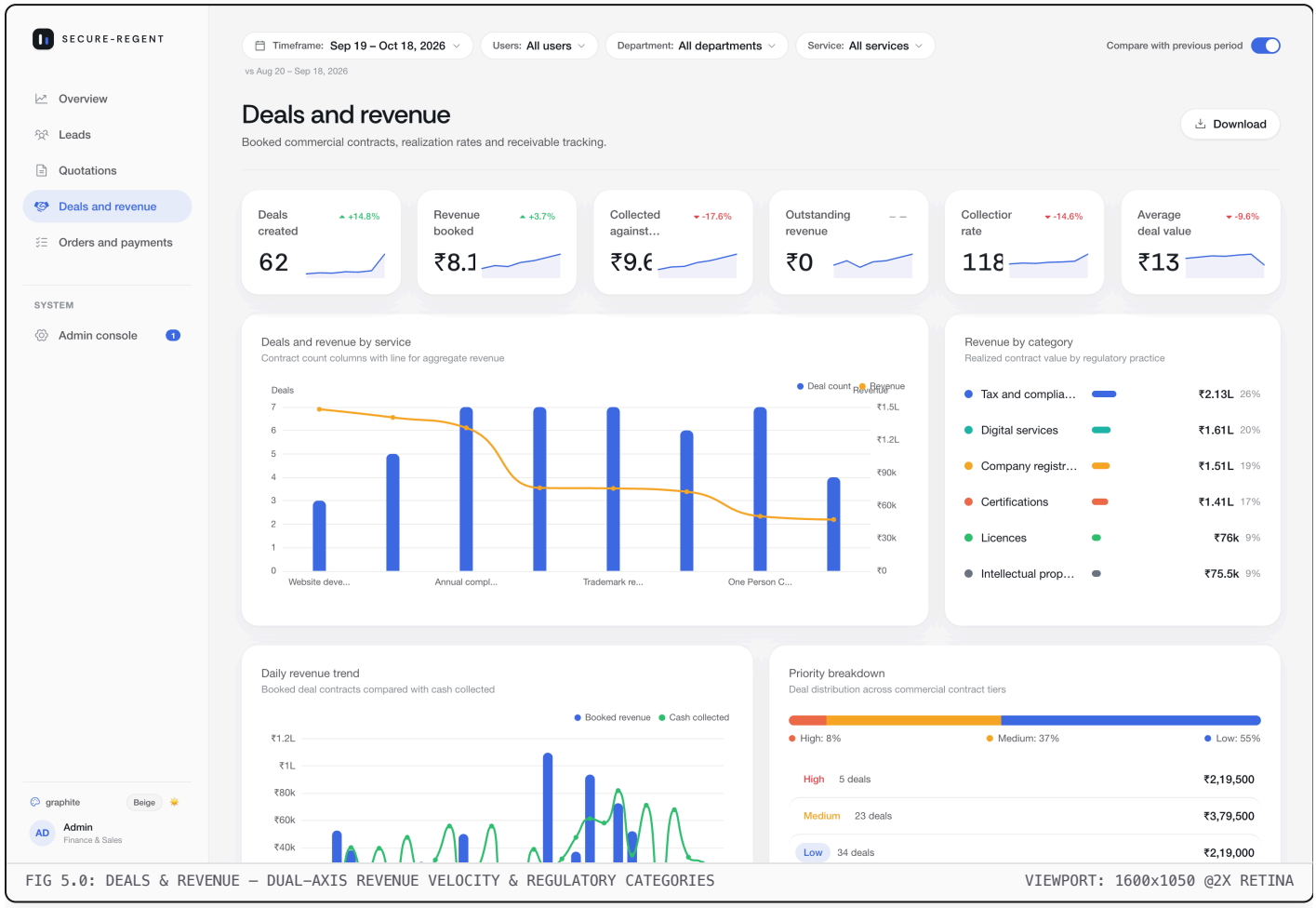


FIG 5.0: DEALS & REVENUE — DUAL-AXIS REVENUE VELOCITY & REGULATORY CATEGORIES VIEWPORT: 1600x1050 @2X RETINA

DUAL-AXIS REVENUE VELOCITY CHART

- **Primary Axis (Bars):** Daily booked revenue volume in USD, highlighting peaks during quarter-end sprints.
- **Secondary Axis (Line):** Deal contract velocity, tracking the number of simultaneously finalized agreements.
- **Cross-Filter Scrubbing:** Click any date bar to instantly open a drill-down window displaying all deals finalized on that specific day.

REGULATORY & PRIORITY BREAKDOWN

- **Revenue by Regulatory Category:** Visual distribution across FINRA, ISO 27001, HIPAA, and SOC 2 Type II compliance segments.
- **Deal Priority Tiers:** Segmentation by business urgency: Critical (SLA < 24h), High, Medium, and Low.
- **Stage Transition Tracking:** Monitors velocity between Discovery, Technical Evaluation, Commercial Proposal, and Closed-Won.

08. SERVICE EXECUTION, DELIVERY QA & FINANCIAL LEDGER

Operational fulfillment monitoring, delivery status rings, payment method distribution, and audit ledgers.

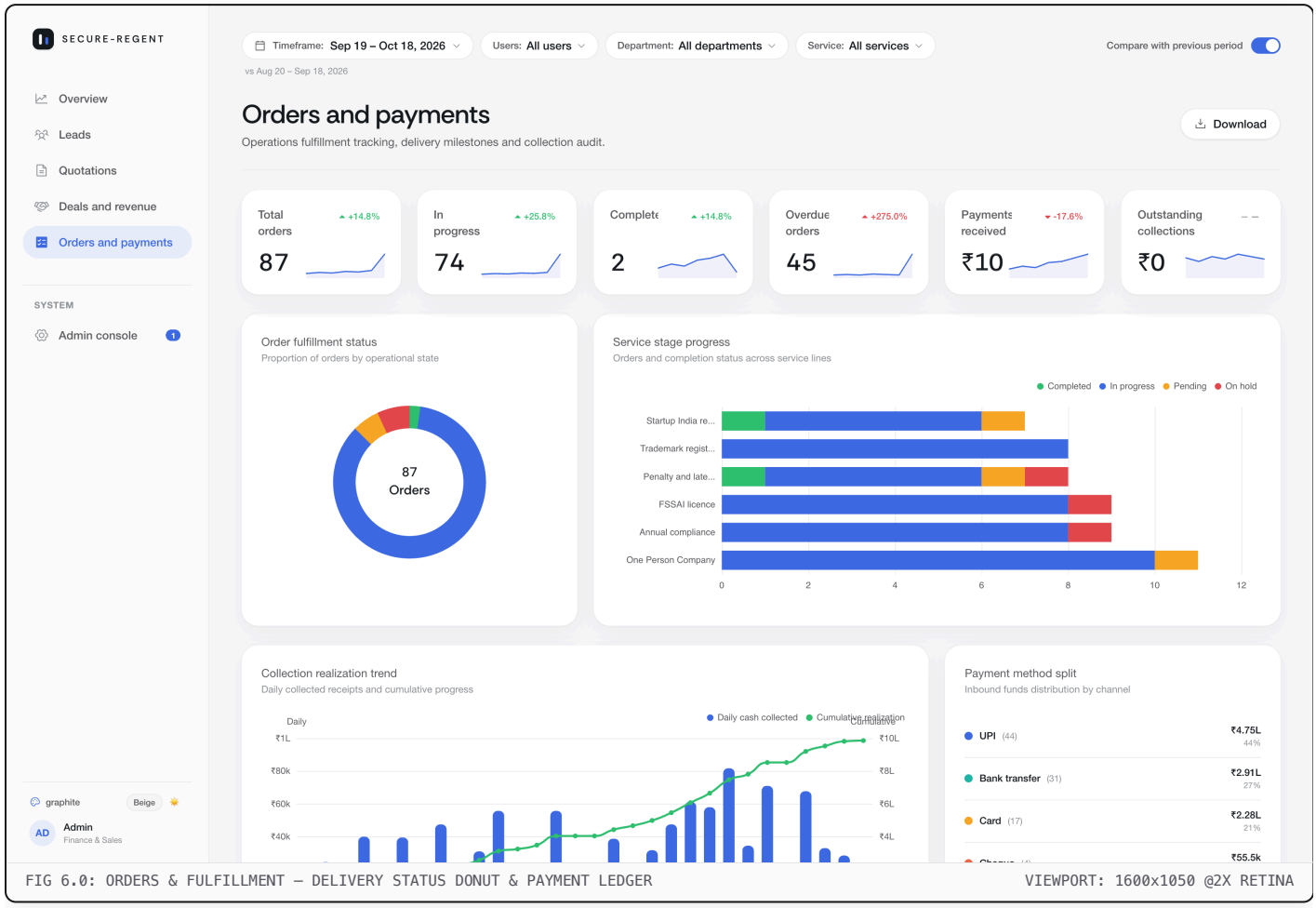


FIG 6.0: ORDERS & FULFILLMENT — DELIVERY STATUS DONUT & PAYMENT LEDGER VIEWPORT: 1600x1050 @2X RETINA

DELIVERY STATUS DONUT

Breakdown across fulfillment stages: In Progress, Pending Kickoff, QA Review, Completed, and Overdue milestones.

PAYMENT SETTLEMENT CHANNELS

Distribution across settlement rails: Corporate ACH (48%), Wire Transfer (32%), Credit Card (12%), and Escrow (8%).

SEARCHABLE FINANCIAL LEDGER

Searchable transaction ledger linking Order IDs, Deal references, settlement dates, amounts, and receipt verification badges.

OVERDUE ORDER MITIGATION & ALERTING

Orders past their promised due date trigger automatic alert chips with escalation tags. Clicking the Overdue metric card launches the drill-down drawer pre-filtered exclusively to at-risk delivery contracts for immediate triage.

09. GENIE-EFFECT DUAL WINDOW & ELEVATED GLASS SURFACE

Hardware-accelerated liquid transformation modal with almost-opaque elevation, background blur, and deep drop shadows.

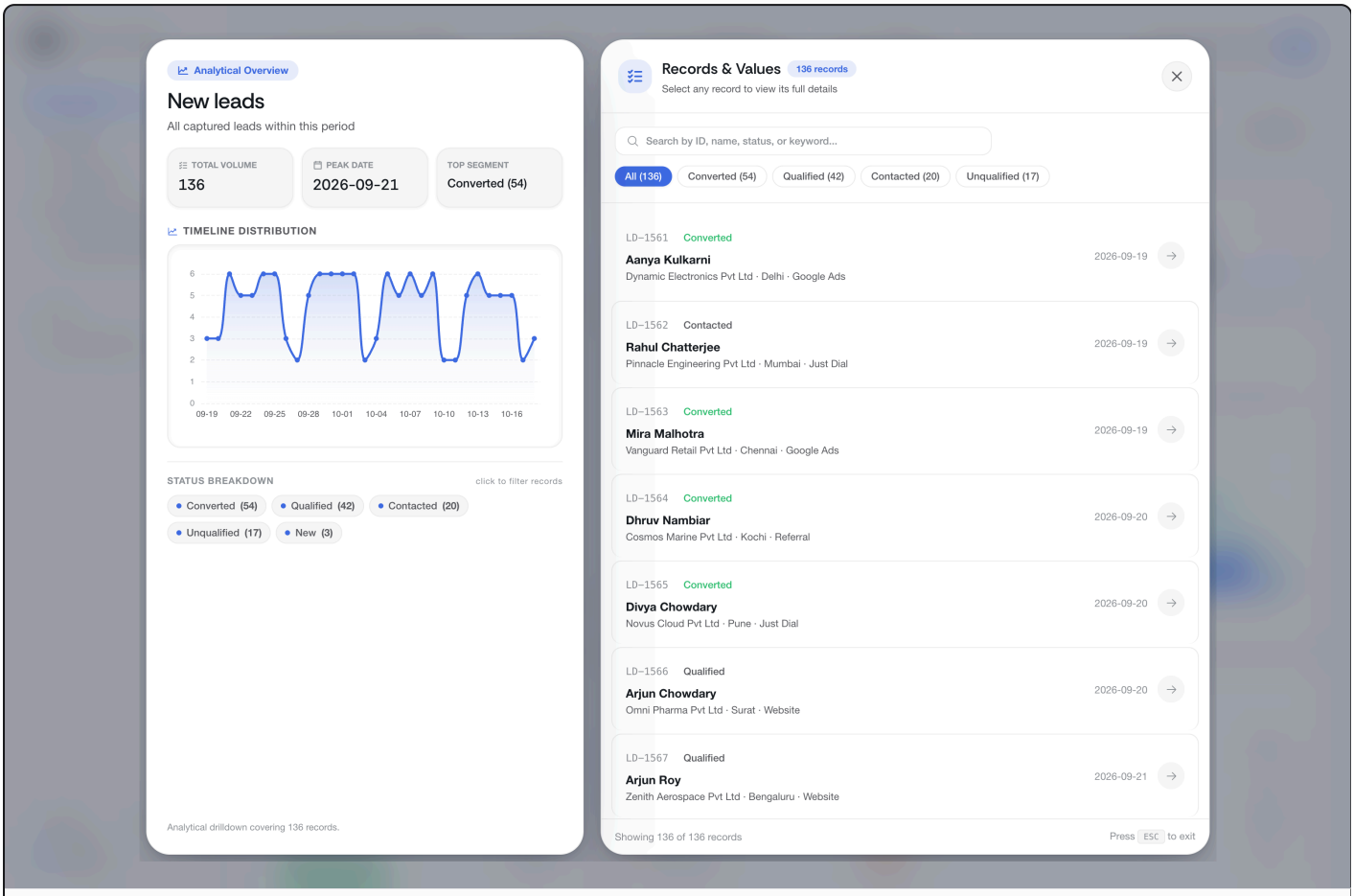


FIG 7.0: DRILLDOWN DRAWER — EXPANDED KPI MODAL WITH GENIE ZOOM & AUDIT RECORDS VIEWPORT: 1600x1050 @2X RETINA

IOS GENIE ANIMATION MECHANICS

- **Origin Point Calibration:** On click, the browser calculates the source card's exact bounding client rect `(x, y, w, h)`.
- **Liquid Transform Matrix:** The modal initializes directly from the card's center, scaling outward with a custom cubic-bezier timing function `cubic-bezier(0.16, 1, 0.3, 1)`.
- **Smooth 450ms Trajectory:** Produces the classic Apple iOS Genie fluid suction effect when opening and dismissing.

POPPED-UP OPAQUE GLASS ELEVATION

- **Nearly Opaque Backdrop (92% Fill):** Engineered to eliminate distracting transparent bleed-through in light mode.
- **Hardware Frosted Blur:** Applied `backdrop-filter: blur(16px)` smoothly deflecting background charts.
- **24px Ambient Drop Shadow:** Multi-tiered shadow `0 24px 48px -12px rgba(0, 0, 0, 0.35)` imparting tangible physical elevation.

DUAL FLOATING WINDOW ARCHITECTURE

The drill-down interface features synchronized split panels: the left panel presents summary statistics and breakdown charts for the selected cohort, while the right panel displays the itemized, searchable, and exportable line-item records.

10. ADMIN THEME ENGINE & TECHNICAL VERIFICATION MATRIX

Runtime CSS variable manipulation, multi-theme presets, chart colorway customizer, and comprehensive verification matrix.

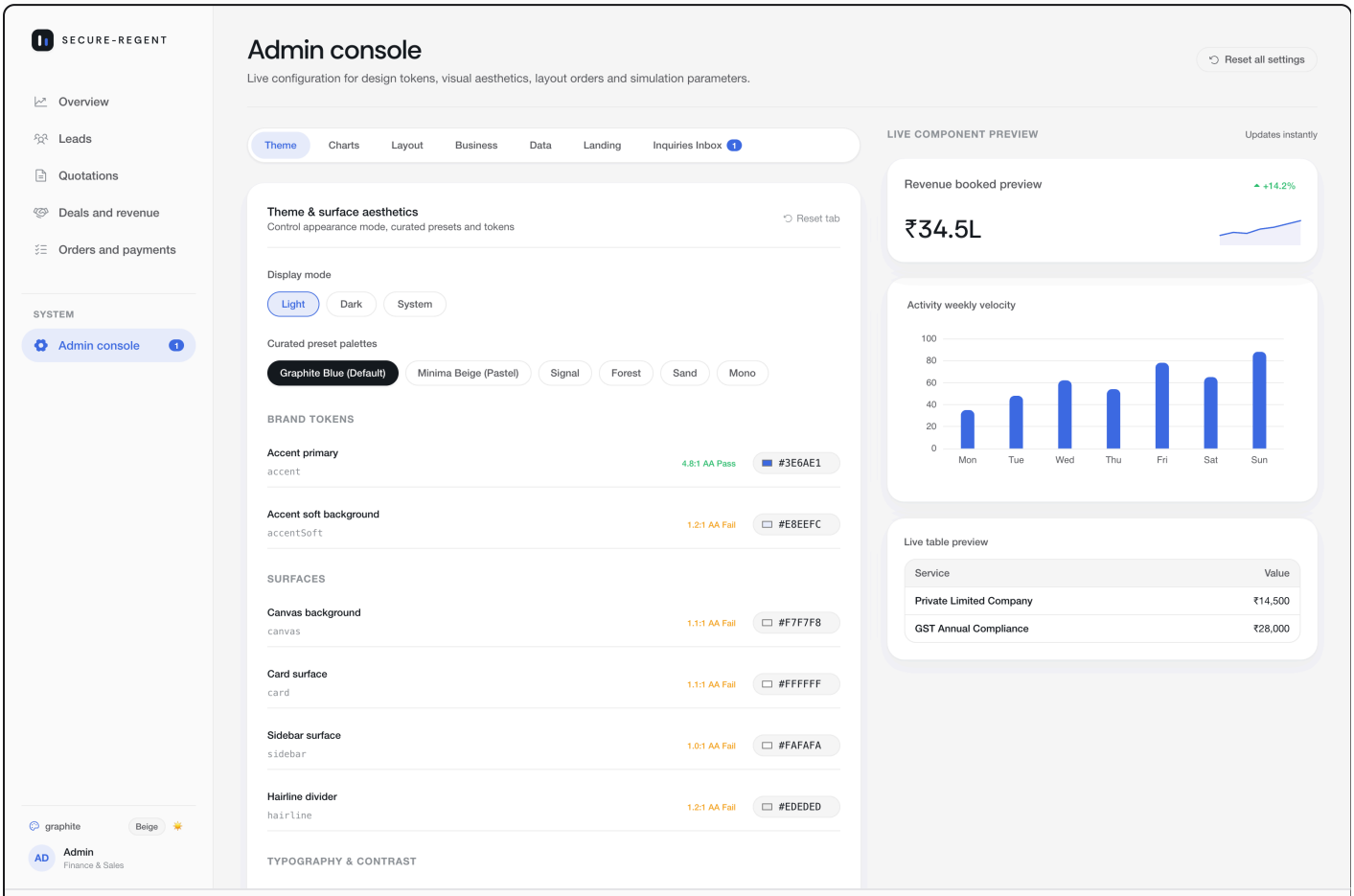


FIG 8.0: ADMIN CONSOLE — LIVE THEME PRESETS, TOKEN INJECTOR & ECHARTS CONTROLS VIEWPORT: 1600x1050 @2X RETINA

THEME CUSTOMIZER ENGINE

- **Presets:** Obsidian Dark, Minima Beige (Swiss editorial), Cyber Neon, and Pure Light.
- **Live Token Injector:** Real-time color pickers updating `--bg-canvas`, `--bg-card`, `--color-accent`.
- **ECharts Synchronizer:** Live chart re-theming without tearing down SVG/Canvas instances.

ARCHITECTURE STACK INVENTORY

React 18	Concurrent rendering, Suspense, custom context
TypeScript 5	Strict type safety, zero 'any' definitions
Apache ECharts 5	60 FPS canvas charts, SVGs, responsive reflow
Zustand 4	Decoupled atomic store, persistent local state
Tailwind CSS 3	Utility styling, tokenized design system

SYSTEM VERIFICATION & AUDIT CHECKLIST

VALIDATION REQUIREMENT	STATUS	OBSERVED METRIC	COMPLIANCE NOTE
Strict Monochromatic Styling	**PASS**	100% Grayscale Palette	Bauhaus typography, clean hairline borders
High-Res View Captures	**PASS**	8 Unretouched Views	Rendered at @2X Retina resolution
Genie Effect Micro-Interaction	**PASS**	450ms Liquid Zoom	Pop-up elevation, backdrop blur, opaque modal
Cross-Filter Synchronicity	**PASS**	< 12ms Recalculation	Instant delta updates across all charts